

PROJECT REPORT: PLANT DISEASE DETECTION VIA CNN

Abstract

This project develops an automated plant disease detection system utilizing a Convolutional Neural Network (CNN). By analyzing leaf images, the model accurately identifies crop diseases. The system integrates a React.js frontend with a FastAPI backend to deliver a scalable, user-friendly corporate solution for agricultural diagnostics.

Introduction

Crop diseases significantly impact agricultural yields and economic stability. Traditional manual inspection is labor-intensive and prone to human error. This project provides a technologically advanced, automated solution. By leveraging deep learning and computer vision, the application allows users to upload leaf images and receive immediate, high-accuracy diagnostic feedback, facilitating early intervention and resource optimization.

Tools Used

- **Machine Learning:** TensorFlow, Keras, NumPy, Pandas.
- **Backend Server:** FastAPI, Uvicorn, Python, Pillow (PIL).
- **Frontend Interface:** React.js, Axios, HTML5, CSS3.
- **Infrastructure:** Git, GitHub, RESTful API architecture.

Steps Involved in Building the Project

1. **Data Engineering:** Acquired and preprocessed a comprehensive dataset of categorized leaf images. Applied normalization and data augmentation techniques to ensure robust model training.
2. **Model Architecture & Training:** Designed a CNN using TensorFlow/Keras. Optimized the network through multiple convolutional and pooling layers, followed by dense layers. Exported the trained model in a deployable format (.pb).
3. **API Development:** Built a backend service using FastAPI to host the trained model. Engineered endpoints to receive image payloads, process them via Pillow, and return diagnostic predictions with confidence scores.
4. **UI/UX Development:** Developed a responsive single-page application (SPA) using React.js, enabling seamless image uploads and clear presentation of diagnostic results.

5. **System Integration:** Connected the frontend and backend architectures. Conducted end-to-end testing with validation images to ensure low-latency performance and high reliability.

Conclusion

The system successfully provides a highly accurate, automated approach to plant disease identification. The integration of a deep learning model within a modern web architecture establishes a robust foundation for scalable agricultural technology solutions. Future iterations will focus on expanding the disease classification repository and optimizing for cloud-based deployment.